

The exact mathematical layout of the SARTA regression polynomial suite and its reference atmospheric architecture provides the framework for its fast calculations.

The Fixed Gas Polynomial Matrix

For well-mixed fixed gases, SARTA calculates the channel optical depth (τ) for an atmospheric layer using a precise **10-term polynomial string**.

Let $\Delta T = T_{\text{actual}} - T_{\text{ref}}$ and $\sec(\theta)$ be the path length modifier. The 10 distinct predictors (P_i) are calculated as follows:

Term (P_i)	Predictor Equation (P_i)	Physical Phenomenon Captured
1	$\sec(\theta)$	Baseline geometric slant path length
2	$\sec^2(\theta)$	High-angle path length correction
3	$\Delta T \cdot \sec(\theta)$	First-order path temperature shift
4	$\Delta T \cdot \sec^2(\theta)$	Non-linear path temperature variation
5	$(\Delta T)^2 \cdot \sec(\theta)$	Symmetric thermal Doppler line-broadening
6	$(\Delta T)^2 \cdot \sec^2(\theta)$	Combined temperature-path non-linearities
7	ΔT	Pure local kinetic temperature variance
8	$(\Delta T)^2$	Pure local thermal broadening offset
9	$\Delta T^3 \cdot$	Higher-order line wing

Term (\$i\$)	Predictor Equation (\$P_i\$)	Physical Phenomenon Captured
	$\sec(\theta)$	asymmetry
10	$\Delta T^4 \cdot \sec(\theta)$	Extremely high-temperature gradient correction

SARTA computes the final effective layer optical depth by calculating the dot product of this string against a pre-compiled array of instrument coefficients (\$A_i\$):

$$\tau_{\text{layer}} = \sum_{i=1}^{10} A_i P_i$$

SARTA Reference Atmospheric Profiles

The reference profile standard (\$T_{ref}\$, \$Q_{ref}\$, \$V_{ref}\$) serves as the baseline from which all polynomial \$\Delta\$ transformations are measured.

- The Baseline Dataset:** SARTA's reference architecture is built upon a globally distributed statistical regression of atmospheric conditions, historically initialized using the **100 standard automated layers** defined by NOAA and the UMBC Atmospheric Spectroscopy Laboratory.
- Pressure Grid Boundaries:** The profile spans from a nominal space boundary near 0.005 hPa down to a standard surface baseline of 1100 hPa . The layers are intentionally structured to be thinner near the surface (high density) to capture complex boundary layer thermodynamics.
- Constructing the Deltas:** When SARTA ingests a real-time weather forecast or data assimilation profile (e.g., from ECMWF or NASA GMAO), it interpolates those measurements onto its fixed 100-layer reference grid. It then extracts the ratio metrics (\$Q/Q_{ref}\$) or differences (\$\Delta T\$) fed into the polynomial strings above.

Would you like to explore how SARTA converts these layer optical depths into final **top-of-atmosphere (TOA) brightness temperatures**, or how it handles **variable trace gases** like \$CO\$ and \$CH_4\$?